

Kalp Shah

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PROFESSIONAL SUMMARY

Software Engineer focused on the intersection of embedded Linux IoT platforms and cloud infrastructure. Experienced in designing, scaling, and automating real-time systems that monitor, control, and optimize power electronics and battery energy storage systems (BESS). Proven expertise across the complete edge-to-cloud pipeline, from low-latency hardware protocol integration (Modbus, CANbus, MQTT) to high-throughput AWS telemetry, Infrastructure as Code, and resilient edge runtimes.

EXPERIENCE

Energy Toolbase

Calgary, AB

Intermediate Software Engineer

June 2023 – Present

- Architect and develop embedded Linux IoT platform software and cloud infrastructure to monitor, automate, and optimize power electronics and battery energy storage systems (BESS) in real time.
- Drive technical design and platform architecture for edge systems, collaborating directly with product stakeholders to translate requirements into resilient, production-ready services.
- Build scalable web APIs and edge-to-cloud transport mechanisms using Python, FastAPI, and Go, interfacing with external hardware devices via Modbus (RTU/TCP), CANbus, Digital I/O, and MQTT.
- Expand telemetry integration with AWS cloud services to ensure high-throughput ingestion pipelines, automated alerting, and low-latency bidirectional device control.
- Streamline deployment consistency across edge and cloud environments using Infrastructure as Code (IaC), SaltStack, Docker, Kubernetes, and CircleCI.
- Ensure platform reliability and security through comprehensive unit/integration testing (TDD/BDD), secure edge-cloud communication protocols, and active production on-call rotation.
- Facilitate code reviews, evaluate emerging technologies, and provide peer mentorship to elevate team-wide engineering standards and architectural best practices.

Software Engineer

Jan. 2021 – June 2023

- Developed and maintained core edge device software services, hardware communication interfaces, and automated test suites for commercial and industrial energy storage deployments.
- Implemented data parsing, state-machine control logic, and local fault mitigation for power conversion systems, improving edge reliability under diverse grid conditions.

Celestica

Newmarket, ON

Test & Automation Developer

May 2018 – Sept. 2019

- Collaborated within a team of 12 senior engineers to design and develop automated test solutions for a tier-1 telecommunications client in the United States.
- Engineered automated test scripts and internal tooling to streamline manufacturing verification processes and improve testing cycle efficiency.
- Built regression test suites and data logging utilities to identify hardware defects early in the production lifecycle.

PROJECTS

Personal Cloud Platform & Infrastructure | *Docker Compose, Cloudflare Tunnels, MySQL, Bash, Python*

- Architected and deployed self-hosted multi-service production infrastructure (kalp.dev) featuring zero-trust ingress via Cloudflare Tunnels.
- Developed automated disaster recovery pipelines with systemd timers, Bash, and offsite storage, validated by Python integration test suites.

Facility Reservation Automation Engine | *Python, Selenium WebDriver, Linux, Raspberry Pi*

- Engineered an automated reservation daemon executing on a 24/7 Raspberry Pi to handle low-latency bookings for high-demand municipal facilities.
- Implemented headless browser scheduling, session state persistence, and automatic retry mechanisms to ensure fault-tolerant booking execution.

EDUCATION

Ryerson University

Toronto, ON

Bachelor of Engineering (B.Eng.) in Computer Software Engineering

TECHNICAL SKILLS

Languages: Python, Go, C/C++, SQL, Bash, JavaScript

IoT & Hardware Protocols: Modbus (RTU/TCP), CANbus, MQTT, Digital I/O, Embedded Linux, Power Electronics, EMS

Cloud & DevOps: AWS, Docker, Kubernetes, SaltStack, CircleCI, Linux, Infrastructure as Code (IaC), Git, CI/CD

Frameworks & Architecture: FastAPI, REST APIs, Microservices, TDD/BDD, Unit & Integration Testing, System Design